

Rachael Lang

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Education

University of Wisconsin–Madison

Master of Science in Computer Science

May 2025

GPA: 3.6

University of Wisconsin–Madison

Bachelor of Science in Computer Science, Minor: Data Science

Dec 2023

GPA: 3.6

Technical Skills

- **Languages:** TypeScript, JavaScript, Java, Python, Kotlin, SQL, C++, Bash
- **Front-End:** React, React Native, HTML, CSS, Material UI, Sass, Responsive & Accessible Design
- **Back-End & Tools:** Spring Boot, Node.js, REST APIs, Microservices, Jest, React Testing Library, Cypress, CI/CD

Work Experience

Target Corporation – Front-End Developer Intern

Jun 2024 – Aug 2024

- Designed and implemented a scalable, task-driven installation workflow using TypeScript and React, leveraging centralized state management to handle dynamic, nested configuration states across multi-step hardware setup processes
- Translated hardware configuration steps into an intuitive, formatted UI with grouped tasks, checkboxes, and clear visual hierarchy, reducing set-up friction by 20%
- Architected reusable, UI components aligned with design systems using Material UI, standardizing patterns across screens and reducing duplicate code by 25% while improving cross-team consistency
- Authored behavior tests using Jest and React Testing Library, increasing front-end test coverage and strengthening regression protection during iterative development

Intel Corporation – Software Engineer Intern

Nov 2023 – Jan 2024

- Engineered a high-performance Python-based fingerprinting system for large directory hierarchies, optimizing recursive traversal through timestamp-based pruning to efficiently scan thousands of files in manufacturing environments
- Designed SQL-backed change detection logic with data integrity safeguards to support concurrent access by over 30 engineers, ensuring reliable identification of new, modified, and deleted configuration states
- Improved scan efficiency by leveraging file modification timestamps to avoid redundant hash computations across large directory trees
- Deployed the tool as a scheduled job during new product introduction cycles, reducing manual debugging time by approximately 30%

Target Corporation – Full-Stack Developer Intern

Jun 2023 – Aug 2023

- Implemented back-end authorization logic in Kotlin and Java within Spring Boot services, handling high-throughput validation of thousands of daily payment transactions with robust error handling and concurrency-safe rule enforcement
- Built and integrated REST API endpoints enabling communication between point-of-sale systems and payment issuers, validating over 100 request/response flows during development
- Designed structured input validation and error handling mechanisms to prevent invalid transaction states and improve reliability of authorization workflows
- Developed SQL queries for transaction data retrieval and rule evaluation, improving clarity and maintainability across back-end services

The J.M. Smucker Company – Software Engineer Intern

May 2022 – Aug 2022

- Led Oracle SQL schema design and migration efforts for a supply planning system, applying normalization strategies and coordinating safe data transformations to support enterprise-scale system modernization with minimal disruption
- Developed an automated Bash-based file transfer solution with logging, retry logic, and compression to reliably handle nightly multi-GB forecast transfers to an external SaaS planning system
- Reduced after-hours manual intervention by 50% by eliminating recurring file transfer failures in automated supply planning workflows

Projects

RachaelTLang.com – Full-Stack Developer

Dec 2023 – Present

- Engineered a high-performance personal web application using HTML, CSS, and JavaScript, achieving 96+ Lighthouse performance scores through asset optimization, structured DOM design, and efficient rendering practices
- Designed a reproducible Git-based deployment workflow using GitHub Pages, enabling automated production releases and streamlined iteration without manual hosting configuration